

VERNADSKY BASE, Antarctica

WMO: 890630

Lat: 65.246S

Lon: 64.258W

Elev: 11

StdP: 101.19

Time Zone: -3.00 (W03)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	-19.3	-16.7	-23.2	0.5	-18.6	-20.1	0.6	-15.7	19.4	0.4	16.8	-1.9	3.3	160	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	2.4	5.1	2.6	4.2	2.1	3.6	1.7	3.3	4.4	2.6	3.6	2.1	3.1	9.1	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2.4	4.5	3.2	1.7	4.3	2.6	1.2	4.1	2.2	15.7	4.6	14.4	3.7	13.5	3.1	5.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.4	14.3	12.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-2.5	1.1	1.2	0.5	-1.6	-2.6	-3.7	-5.4	-6.8	-5.7	-3.6	-2.1	-0.3
	DBStd	4.05	1.28	1.45	1.58	2.21	2.56	2.70	3.72	4.75	5.35	3.87	2.84	1.33
	HDD10.0	4546	276	246	296	349	389	412	479	522	472	422	363	319
	HDD18.3	7587	534	479	554	599	647	662	737	781	722	681	613	578
	CDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	4.4	2.6	4.0	4.9	4.5	4.5	5.1	4.8	4.5	5.5	4.7	4.2	2.9
Precipitation	PrecAvg	763	61	63	75	64	62	52	56	59	69	79	71	53
	PrecMax	995	112	102	112	115	115	90	101	90	98	130	115	120
	PrecMin	546	22	39	41	28	37	26	22	20	38	44	36	17
	PrecStd	124	22	15	14	18	18	16	18	18	15	20	23	22
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.0	6.7	5.7	4.5	5.4	2.9	2.5	3.5	3.1	4.4	4.1	4.5
		MCWB	3.3	3.3	3.1	2.7	2.2	0.5	-0.4	0.9	1.0	2.2	2.3	1.5
	2%	DB	4.7	4.7	4.3	3.5	3.4	1.7	1.3	1.8	1.9	3.0	2.5	3.2
		MCWB	2.5	2.7	2.0	1.6	1.0	-0.2	-0.6	0.2	0.3	1.3	1.1	1.1
	5%	DB	3.7	3.8	3.4	2.3	1.9	0.6	0.4	0.4	1.1	2.0	1.7	2.4
		MCWB	1.9	2.1	1.6	1.0	0.2	-0.8	-1.0	-0.9	-0.1	0.6	0.5	0.8
	10%	DB	2.9	3.2	2.6	1.4	0.8	-0.3	-0.4	-0.8	0.4	1.1	1.0	1.7
		MCWB	1.4	1.8	1.2	0.2	-0.3	-1.4	-1.5	-2.0	-0.5	0.0	0.0	0.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.2	4.1	3.5	3.3	2.8	0.7	0.5	1.2	1.7	3.1	2.5	2.6
		MCDB	5.3	5.3	5.0	4.1	5.7	2.3	1.3	2.8	2.3	3.9	3.2	3.4
	2%	WB	3.0	3.2	2.5	2.0	1.2	0.1	-0.2	0.2	0.8	1.6	1.5	1.8
		MCDB	4.0	4.3	3.7	3.0	2.8	1.3	0.7	1.3	1.4	2.5	2.2	2.5
	5%	WB	2.2	2.6	2.0	1.2	0.4	-0.5	-0.8	-0.6	0.2	0.8	0.8	1.2
		MCDB	3.1	3.6	3.0	2.2	1.6	0.5	0.2	0.3	0.9	1.8	1.4	2.0
	10%	WB	1.8	2.1	1.4	0.4	-0.2	-1.1	-1.5	-1.9	-0.4	0.1	0.3	0.6
		MCDB	2.6	2.9	2.3	1.3	0.7	-0.3	-0.5	-1.2	0.4	0.9	0.8	1.4
Mean Daily Temperature Range	5% DB	MDBR	2.5	2.4	2.4	2.3	2.7	3.1	3.7	4.5	5.4	4.3	3.8	2.9
		MCDBR	3.8	3.4	3.9	3.4	4.6	3.7	4.3	4.7	4.8	4.2	3.9	4.1
		MCWBR	2.6	2.2	2.6	2.8	2.8	2.7	3.7	3.5	4.2	3.1	3.1	2.8
	5% WB	MCDBR	3.4	3.2	3.5	3.2	4.2	3.5	4.1	4.8	5.1	4.0	3.7	3.5
		MCWBR	2.5	2.3	2.6	2.8	2.9	2.8	3.8	3.8	4.9	3.3	3.1	2.8
Clear-Sky Solar Irradiance	taub		0.284	0.281	0.266	0.237	N/A	N/A	N/A	0.233	0.258	0.276	0.279	0.284
	taud		2.383	2.402	2.412	2.311	N/A	N/A	N/A	2.161	2.266	2.258	2.303	2.342
	Ebn at Noon		980	928	846	665	N/A	N/A	N/A	633	829	919	977	994
	Edh at Noon		105	91	70	45	N/A	N/A	N/A	50	78	104	113	113
All-Sky Solar Radiation	RadAvg		6.07	4.08	2.15	0.76	0.16	0.03	0.09	0.52	1.77	3.94	6.03	7.02
	RadStd		0.36	0.25	0.15	0.06	0.01	0.00	0.01	0.04	0.16	0.23	0.38	0.57

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page